

How do musical attributes impact the popularity of songs on Spotify from 2015 to the present?

Research Question

“ How do musical attributes—[danceability, energy, key, loudness, acoustic-ness, valence, tempo, mode, and duration]—impact the popularity of songs on Spotify from 2015 to the present? “

Why Important?

By identifying specific musical attributes the music industry can make their music align trendier for commercial success.

Data Collection

Data:

“10000 Spotify Songs 1960-Now”

- Originally collected from Spotify’s API
- Popular songs based on ARIA (Australian Recording Industry Association) and Billboard charts rankings.
- Vetted by Kaggle, with a usability score of 10.0 and licensed under CC0: Public Domain.
- The dataset is curated from reliable sources, ensuring the data is not simulated.

Variables:

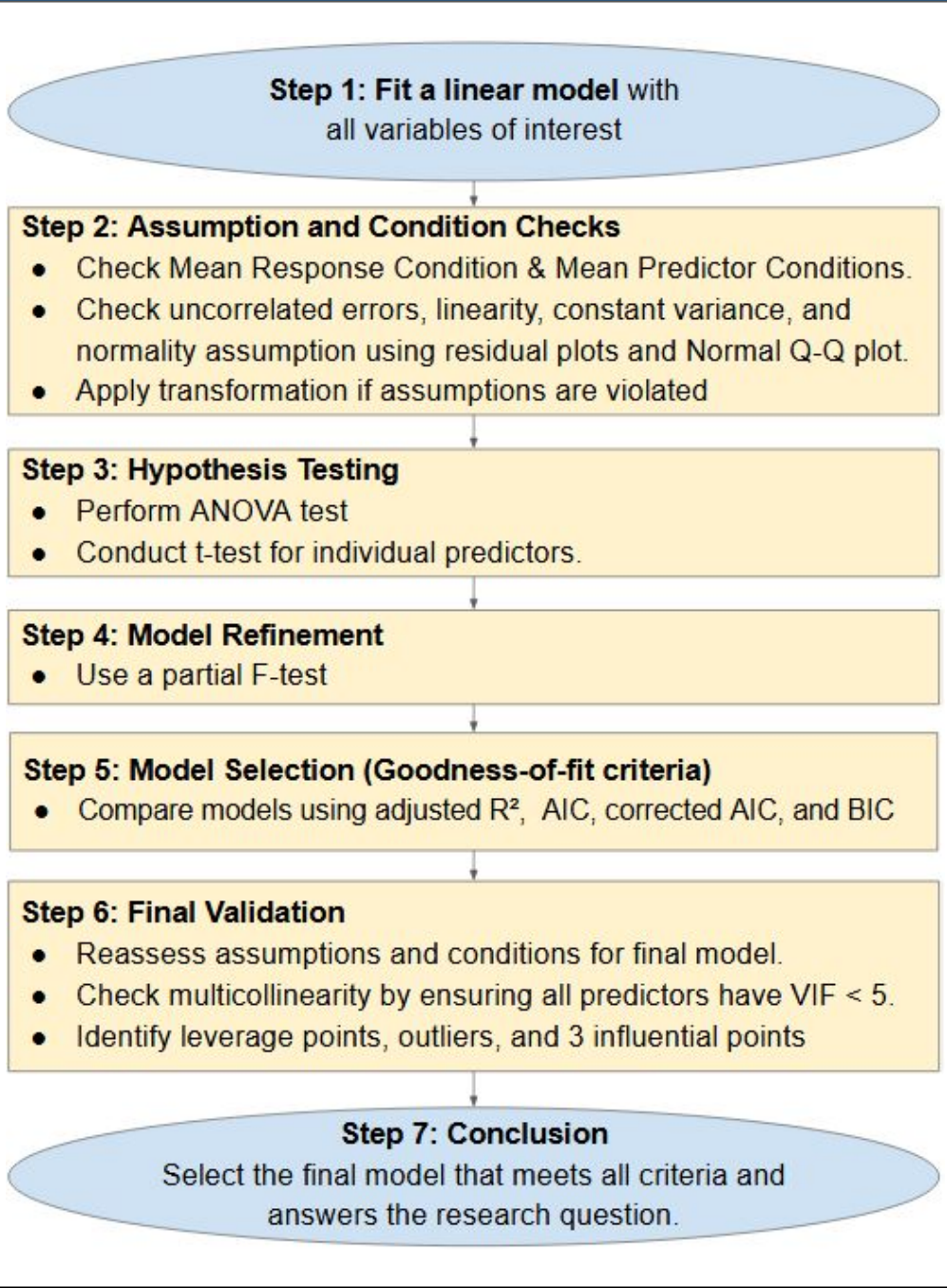
Response Variable
Popularity
Predictor Variables
Danceability, Energy, Key, Loudness, Acoustic-ness, Valence, Tempo, Mode, and Duration

These variables are important for answering our research question.

Limitations

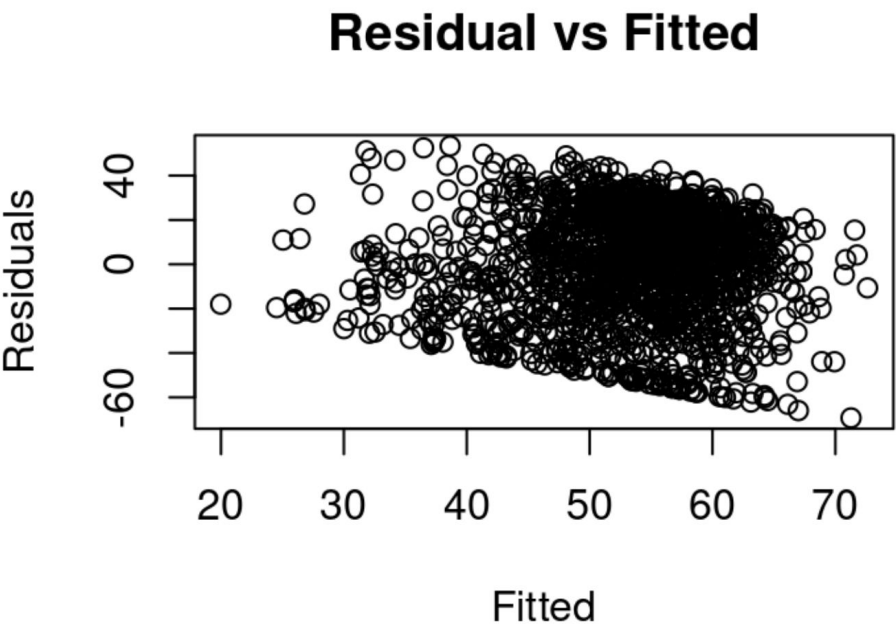
- Low R-squared value:** predictors explain only a small portion of the variation in song popularity.
- Exclusion of songs with a popularity score of 0** may introduce bias.
- Outliers** (e.g., observations 5 and 7) may distort coefficient estimates and predictions.
- Slight violation of normality assumption** due to left-skewness in popularity data, impacting p-values and test reliability.

Methods



Results

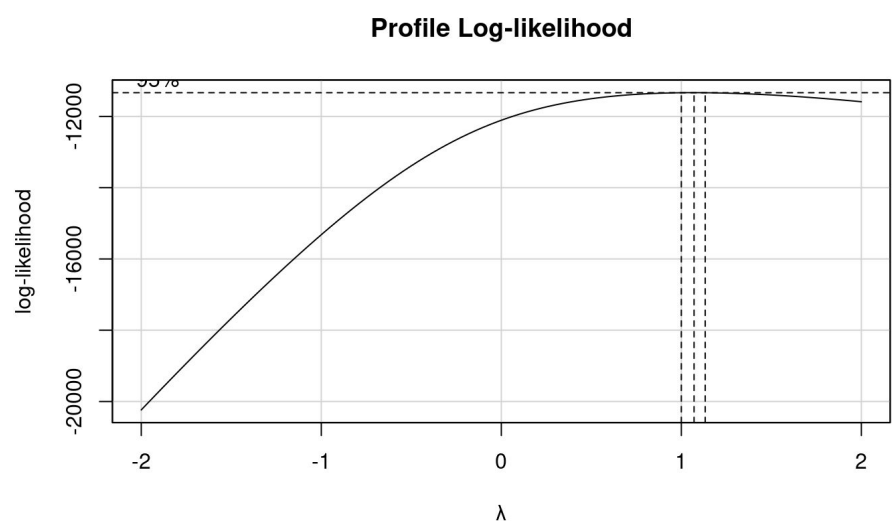
1. Residual Plot



Most data points appear randomly scattered and unclustered.

- Supports linearity, constant variance, uncorrelated.

2. Confidence Interval for Response



95% Confidence Interval of MLE for Popularity overlaps with 1

- Transformation is not necessary for the variable Population.

3. Power Transformations to Multicollinearity

Variables	Rounded Power	Transformed Variables
Danceability	1.65	Danceability ^(8/5)
Energy	1.42	Energy ^(7/5)
Tempo	0.00	ln(Tempo)
Valence	0.76	Valence ^(3/4)
Acousticness	0.28	Acousticness ^(1/2)
Duration	0.00	ln(Duration)

- Danceability, Energy, Valence, and Acousticness were power-transformed with λ = 1.65, 1.42, 0.76, 0.28 respectively.

4. T-test for Individual Coefficient

Variables with coefficient intervals including 0 in the T-test, such as **Key**, **Mode** and **Duration**, are *omitted* as they are considered insignificant.

Reduced Model

```
reduced <- lm(Popularity ~
  bcDanceability + Loudness +
  bcEnergy + bcValence +lnDuration,
  data = new)
```

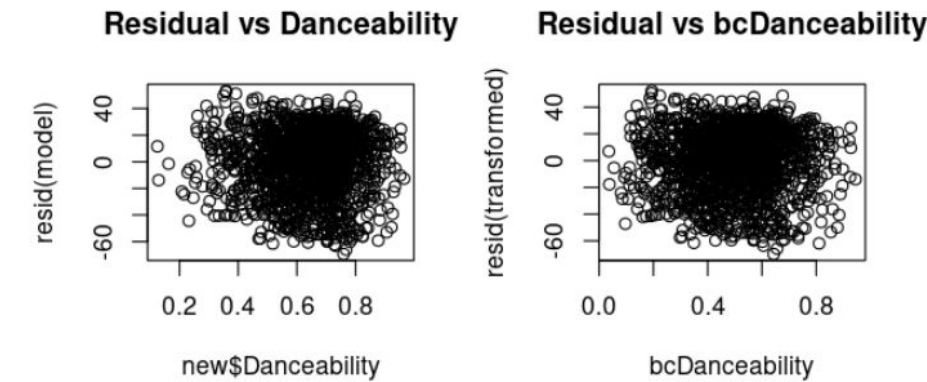
5. Partial F-test

- Reduced model is preferable, as the F-statistic is not significant
- Power-transformed Energy (bcEnergy) has low p-value of 0.2427, so we *drop* the variable **bcEnergy**.

6. Improvements

1 < Variance inflation factors (VIFs) < 5

- No severe multicollinearity is present.



- Skewness of residual plot is improved.

Final Model

Response Variable
Popularity
Predictor Variables
bcDanceability, Loudness, bcValence, lnDuration

The analysis reveals that **Danceability**, **Loudness**, **Duration**, and **Valence** are significant predictors of Spotify song popularity.

Danceability, loudness and duration positively impact popularity, while valence decrease it, suggesting that neutral or somber tones may resonate more. Other attributes such as key, mode, energy, acousticness, and tempo were found to have minimal influence.